Econometrics II – Assignment 3:
A Dynamic Model for U.S. House Prices

Spring 2026

1 Introduction

Housing is the largest asset on the balance sheet of most U.S. households and house-price swings have, historically, propagated to the wider macroeconomy; the user-cost framework predicts that, in equilibrium, prices are tied to rents and to the mortgage rate. This report tests that prediction by estimating the long-run relation

$$h_t = \alpha + \beta_1 r_t + \beta_2 m_t, \tag{1}$$

where h_t is log house prices, r_t is log rent, and m_t is the 30-year fixed mortgage rate. Using monthly data for 2012M1–2023M12, we estimate a CVAR with $k = 4$ lags and an unrestricted constant by Gaussian maximum likelihood, test the strict user-cost restriction $\beta_1 = 1$ by a Johansen LR test, and check the sensitivity of the conclusion to extending the sample back to 2000M1.

We find one co-integrating relation, but a rent elasticity statistically above one and a mortgage coefficient indistinguishable from zero, so the strict user-cost benchmark is rejected; equilibrium adjustment is borne almost entirely by the mortgage rate, and the finding is sensitive to the inclusion of the 2008-09 crisis years.

2 Description of the Data

The data come from `Assignment_3.xlsx` and are based on FRED series for the S&P/Case-Shiller U.S. National Home Price Index, the CPI for rent of primary residence, and the 30-year fixed mortgage rate. The transformed variables are $h_t = \log(\text{HousePrice})$, $r_t = \log(\text{Rent})$, and $m_t = \text{MortgageRate}$. The effective sample contains 144 monthly observations.

Figure 1 shows that both h_t and r_t trend upward, while m_t is much more cyclical and rises sharply in 2022–2023. The visual pattern is consistent with non-stationary behaviour and motivates a co-integration analysis rather than a stationary VAR in levels. Table 1 reports basic moments of the three series.

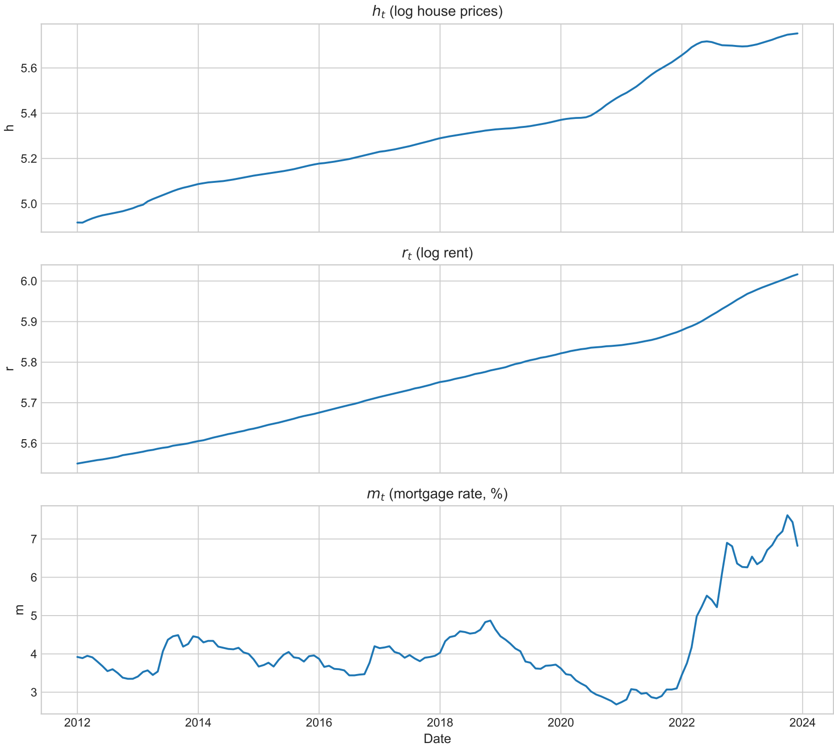

Figure 1: Monthly series for h_t, r_t, and m_t, 2012M1–2023M12.

Table 1: Summary statistics for the analysis variables, 2012M1–2023M12.

Variable	Mean	Std. Dev.	Min	Max
h	5.3108	0.2402	4.9165	5.7527
r	5.7511	0.1287	5.5501	6.0165
m	4.1642	1.0691	2.6800	7.6200

3 Econometric Theory

ADF testing. For each variable y_t we run an augmented Dickey–Fuller regression $\Delta y_t = (\rho - 1)y_{t-1} + \sum_{j=1}^{p} \phi_j \Delta y_{t-j} + c + \varepsilon_t$ and test $H_0 : \rho = 1$ against $\rho < 1$. The statistic is the t-ratio on $(\rho - 1)$ and its asymptotic distribution is the non-standard Dickey–Fuller distribution (not $N(0, 1)$); we use the constant-only critical values throughout. Lag length is selected by AIC.

The CVAR. Let $X_t = (h_t, r_t, m_t)'$. Since the three series are treated as I(1), the natural framework is a co-integrated vector autoregression in error-correction form,

$$\Delta X_t = \Pi X_{t-1} + \Gamma_1 \Delta X_{t-1} + \cdots + \Gamma_{k-1} \Delta X_{t-k+1} + \mu + e_t, \tag{2}$$

where Π contains long-run information, Γ_j capture short-run dynamics, and e_t is a zero-mean innovation. If Π has reduced rank r, we can factorize $\Pi = \alpha\beta'$, so

$$\Delta X_t = \alpha\beta' X_{t-1} + \Gamma_1 \Delta X_{t-1} + \cdots + \Gamma_{k-1} \Delta X_{t-k+1} + \mu + e_t. \tag{3}$$

Here β contains the co-integrating vectors and $\alpha_h, \alpha_r, \alpha_m$ measure how house prices, rents, and the mortgage rate respectively respond to disequilibrium in the long-run relation. The constant is unrestricted, so the long-run relation may include a non-zero intercept. We estimate (3) by Gaussian maximum likelihood, which requires (i) correct lag length, (ii) i.i.d. Gaussian innovations, and (iii) constant parameters; rank inference is robust to non-normality, but inference on individual coefficients relies on the latter two assumptions.

Rank determination. If $r = 0$ there is no co-integration and the model reduces to a VAR in differences; if $0 < r < p$ there are r stationary long-run relations; if $r = p$ the system is stationary in levels. We test $H_0 : \text{rank}(\Pi) \leq r_0$ against $H_1 : \text{rank}(\Pi) > r_0$ by the Johansen trace statistic

$$\text{LR}_{\text{trace}}(r_0) = -T \sum_{i=r_0+1}^{p} \ln(1 - \hat{\lambda}_i),$$

where $\hat{\lambda}_i$ are the ordered eigenvalues from the reduced-rank regression. The trace statistic has a non-standard asymptotic distribution depending on the deterministic specification; we use the case of an unrestricted constant in the VECM and the corresponding Johansen (1995) critical values. The procedure is sequential, starting at $r_0 = 0$ and stopping at the first non-rejection.

Restrictions on β. A linear restriction of the form $\beta = H\phi$ (equivalently $R'\beta = 0$ for $R = H_\perp$) is tested by the LR statistic

$$\text{LR} = -T \sum_{i=1}^{r} \Big[\ln(1 - \tilde{\lambda}_i) - \ln(1 - \hat{\lambda}_i)\Big] \xrightarrow{d} \chi^2(s),$$

where $\tilde{\lambda}_i$ are the restricted eigenvalues and s is the number of restrictions. We use this to test the strict user-cost restriction $\beta_r = 1$, in which case $H = \big((1, -1, 0)', (0, 0, 1)'\big)$ and $s = 1$. Although $\hat{\beta}$ is super-consistent and mixed-Gaussian (while $\hat{\alpha}$ is $\sqrt{T}$-Gaussian), the LR statistic for smooth restrictions of this form is still asymptotically $\chi^2(s)$ under standard regularity. An analogous LR test of $\alpha = A\psi$ (a restriction on the loadings) has distribution $\chi^2(r \cdot (p - m))$, with $m = \dim(\psi)$; in particular, weak exogeneity of variable i for β corresponds to setting the i-th row of α to zero.

4 Empirical Analysis

4.1 Integration Properties

Table 2 reports ADF tests. The level series are clearly non-stationary: the null of a unit root is not rejected for h_t, r_t, or m_t. In first differences the null is rejected for h_t and m_t at the 5% level, while Δr_t is borderline ($p = 0.087$), which is not unusual in a short monthly sample with strong persistence in rents. Taken together, the evidence is consistent with treating the three series as I(1) and proceeding to co-integration testing.

Table 2: ADF unit-root tests with constant; lag length chosen by AIC.

Variable	Stat. (lvl)	p (lvl)	Lags (lvl)	Stat. (Δ)	p (Δ)	Lags (Δ)
h_t	0.2679	0.9758	14	-3.2160	0.0191	1
r_t	1.7251	0.9982	12	-2.6296	0.0870	8
m_t	-1.6255	0.4698	5	-3.6555	0.0048	4

4.2 Lag Selection

Lag length is chosen by estimating a VAR in levels and minimising AIC; Table 3 shows that AIC is minimised at lag 4, with BIC and HQIC also pointing to lag 4 within the displayed range. The preferred VECM therefore uses $k_\Delta = 3$ lagged differences.

Table 3: Lag-order selection criteria for VAR in levels (2012M1–2023M12).

Lag	AIC	BIC	HQIC
0	-10.3391	-10.2736	-10.3124
1	-29.0146	-28.7525	-28.9081
2	-30.7035	-30.2449	-30.5171
3	-30.9077	-30.2525	-30.6414
4	-31.1498	-30.2980	-30.8037
5	-31.1138	-30.0655	-30.6878
6	-31.0758	-29.8309	-30.5699
7	-31.0536	-29.6122	-30.4679
8	-31.0042	-29.3662	-30.3386
9	-31.0864	-29.2519	-30.3409
10	-31.0830	-29.0519	-30.2577
11	-31.0071	-28.7795	-30.1019
12	-30.9735	-28.5494	-29.9885

4.3 Co-Integration Rank

With the dynamic specification fixed, we test the co-integration rank. Table 4 reports the Johansen trace statistic at each r_0. For $H_0 : r = 0$ against $H_1 : r \geq 1$, the statistic of 48.23 exceeds the 5% critical value of 29.80, so we reject. For $H_0 : r \leq 1$ against $H_1 : r \geq 2$, the statistic of 11.56 is below the 5% critical value of 15.49, so we do not reject. The preferred rank is therefore $r = 1$.

Table 4: Johansen trace test. Sequential procedure rejects if trace stat. exceeds the 5% critical value.

H_0	Trace stat.	CV 90%	CV 95%	CV 99%
$r = 0$	48.2266	27.0669	29.7961	35.4628
$r \leq 1$	11.5648	13.4294	15.4943	19.9349
$r \leq 2$	3.6207	2.7055	3.8415	6.6349

Having found rank $r = 1$, we estimate the CVAR and characterise the long-run relation. Figure 2 plots the actual h_t together with the fitted long-run component implied by the normalised co-integrating vector, and Table 5 reports the implied coefficients with ML standard errors.

Table 5: Preferred CVAR (k=4, unrestricted constant, rank 1). Normalised on h_t. Std. err. from VECM ML.

Parameter	Estimate	Std. err.
β_r (rent elasticity)	1.4918	0.0467
β_m (mortgage coef.)	0.0085	0.0068
α_h	0.0156	0.0049
α_r	0.0091	0.0018
α_m	0.3958	0.5380

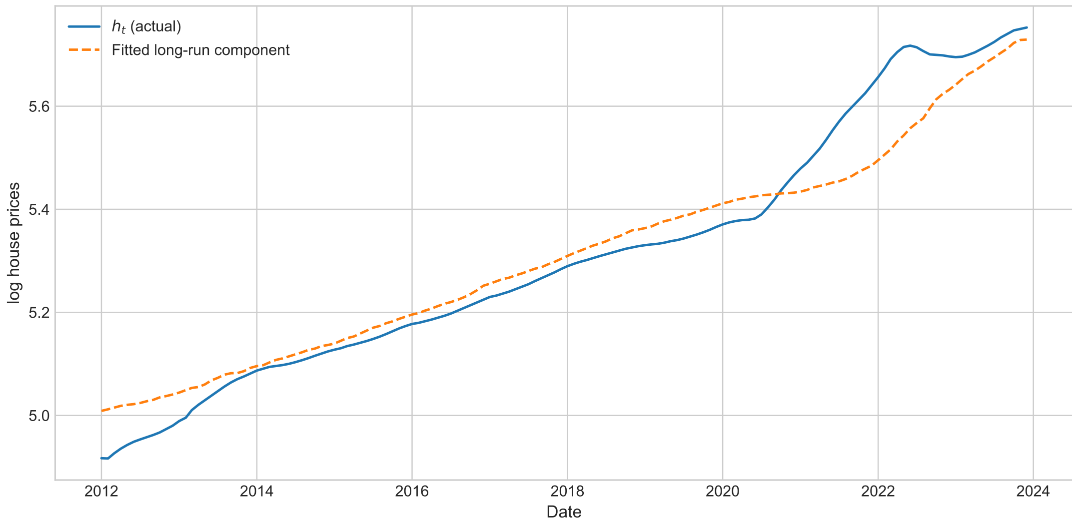

Figure 2: Actual h_t and fitted long-run component from the preferred CVAR.

4.4 Misspecification Diagnostics

Before testing the user-cost restriction, we assess whether the dynamic specification is adequate. Residual diagnostics in Table 6 are mixed. The Portmanteau whiteness test ($p = 0.30$) finds no remaining serial correlation, so the dynamic specification is statistically adequate in that dimension. The Jarque–Bera test strongly rejects normality, and ARCH-LM rejects homoskedasticity for the mortgage-rate residuals at the 5% level ($p = 0.022$) and is borderline for h_t ($p = 0.086$). These results support consistency of the rank estimate (which does not rely on Gaussianity), but the violations of normality and homoskedasticity imply that the standard errors on β and α should be read with caution and that asymptotic-normality based tests are at best approximate; the post-2021 interest-rate shock is the likely driver. The LR statistics reported below remain asymptotically χ^2 under the maintained assumptions, but their finite-sample p-values should be treated as approximate.

Table 6: Misspecification diagnostics for the preferred CVAR specification.

Test	Statistic	p-value
Whiteness (Portmanteau)	83.9923	0.3012
Normality (Jarque–Bera)	391.7401	0.0000
ARCH LM (h)	19.1037	0.0861
ARCH LM (r)	6.4636	0.8909
ARCH LM (m)	23.7053	0.0223

4.5 Preferred CVAR and the User-Cost Restriction

Normalising on house prices, the estimated long-run relation is

$$h_t = \hat{c} + 1.4918\, r_t + 0.0085\, m_t, \tag{4}$$

with ML standard errors of 0.0467 on the rent coefficient and 0.0068 on the mortgage-rate coefficient. The rent elasticity is statistically and economically larger than one (a z-ratio of $(1.4918 - 1)/0.0467 \approx 10.5$ rejects $\beta_r = 1$ at any conventional level), and the mortgage coefficient is small, statistically insignificant ($t \approx 1.25$), and has the wrong sign relative to the user-cost prediction $\beta_2 < 0$. The formal Johansen LR test of the strict user-cost restriction $\beta_r = 1$ in Table 7 corroborates this: the LR statistic of 14.46 has a $\chi^2(1)$ p-value of 0.0001, so we strongly reject unit rent elasticity. The data support a stable long-run link between house prices, rents, and the mortgage rate, but they reject the exact sign and magnitude pattern of the textbook model.

The adjustment loadings indicate how equilibrium is restored. The point estimate $\hat{\alpha}_m = 0.3958$ looks large but has standard error 0.5380, so a t-ratio of 0.74 means we cannot distinguish α_m from zero; the mortgage equation's apparent correction is not statistically supported. The loadings $\hat{\alpha}_h = 0.0156$ ($t \approx 3.2$) and $\hat{\alpha}_r = 0.0091$ ($t \approx 5.0$) are statistically significant but economically tiny. To make this precise we formally test weak exogeneity of the mortgage rate for β, i.e. the restriction $\alpha_m = 0$, using the Johansen LR procedure on $\alpha = A\psi$ with $A = \big((1,0,0)', (0,1,0)'\big)$. Table 7 reports an LR statistic of 0.45 with $\chi^2(1)$ p-value 0.50, so we do not reject weak exogeneity of m_t. The economic picture is therefore that short-run correction is borne by the two slow-moving prices (h_t and r_t), with the mortgage rate behaving as the (weakly) exogenous driver of the system — which is again the opposite of the user-cost story, in which house prices adjust to the rent–cost wedge.

Table 7: Johansen LR tests of restrictions on the long-run structure. $\beta_r = 1$ imposes unit rent elasticity; $\alpha_m = 0$ imposes weak exogeneity of the mortgage rate for β. Both have asymptotic distribution $\chi^2(1)$.

Test	Statistic	d.f.	p-value
LR test $\beta_r = 1$	14.4616	1	0.0001
LR test $\alpha_m = 0$	0.4497	1	0.5025

5 Discussion

The empirical evidence supports one co-integrating relation between house prices, rents, and the mortgage rate in 2012M1–2023M12, but the economic content of the relation is only partly aligned with the textbook user-cost model: the rent elasticity is significantly above one, the mortgage-rate coefficient has the wrong sign and is insignificant, and the formal LR test of $\beta_r = 1$ is rejected at $p = 0.0001$. Simpler frictions absent from (3) — risk premia, refinancing constraints, supply rigidities, expectational shifts — offer natural explanations for the elasticity exceeding one. Equilibrium is also restored in a way that the textbook does not predict: the mortgage rate, not house prices, does most of the short-run adjusting toward the long-run wedge.

Table 8: Robustness of the main results to extending the sample to 2000M1–2023M12.

Quantity	2012M1–2023M12	2000M1–2023M12
VAR lag (AIC)	4	4
Trace stat. at $r = 0$	48.23	19.64
Trace stat. at $r \leq 1$	11.56	3.68
β_r	1.4918	1.5521
β_m	0.0085	0.1719
LR stat. ($\beta_r = 1$)	14.4616	2.7022
LR p-value	0.0001	0.1002

To gauge robustness we re-estimate the same specification on the longer sample 2000M1–2023M12, which includes the housing boom and the financial crisis. AIC again selects four lags, and the implied long-run coefficients ($\hat{\beta}_r = 1.55$, $\hat{\beta}_m = 0.17$) are qualitatively similar. The trace statistic at $r = 0$ collapses to 19.6, however — below the 5% critical value of 29.8 — so we cannot even reject no co-integration on the longer sample, and the LR test of $\beta_r = 1$ has $p = 0.10$, no longer significant at the 5% level. The 2012–2023 finding is therefore sample-specific: the bubble–bust period is informative enough about the data-generating process that the post-crisis "stable equilibrium" interpretation does not survive its inclusion. This sensitivity, combined with the residual diagnostics, suggests the estimated long-run relation should be read as a useful approximation for the post-crisis regime rather than a structural law.

6 Conclusion

For 2012M1–2023M12 a CVAR with $k = 4$ lags identifies one co-integrating relation among house prices, rents, and the mortgage rate. The strict user-cost benchmark is rejected: $\hat{\beta}_r = 1.49$ with $p < 10^{-4}$ for $H_0 : \beta_r = 1$, and the mortgage-rate coefficient is small, insignificant, and of the wrong sign. Equilibrium is restored mainly through the mortgage-rate equation, and the finding does not survive extension of the sample to include the 2000s boom and the 2008-09 crisis.

References

Anundsen, A.K. (2019). Detecting Imbalances in House Prices: What Goes Up Must Come Down? *The Scandinavian Journal of Economics*, 121(4), 1587–1619.

Johansen, S. (1995). *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press.